

Suanne Sparrow | Mechatronics Engineer

(709) 771-6749 | suannesparrow@gmail.com | [LinkedIn](#) | [Portfolio](#)

Education

University of British Columbia

Bachelor of Applied Science, Manufacturing Engineering

Sep 2021 – May 2026

National University of Singapore

Exchange Semester, College of Design and Engineering, Mechanical Engineering

Jan 2025 – May 2025

Experience

Systems Engineering Intern

Creative Energy

Sep 2024 – Dec 2024

Vancouver, BC

- Built a dynamic process simulator (**digital twin**) in **DeltaV Distributed Control Systems** for Canada's largest net-zero development serving **6,000+** residential units
- Owned end-to-end development of a **4-level HMI** compliant with **ISA-101** standard to streamline workflow
- Tuned **PID controllers** and performed **V&V of closed-loop control strategies** to optimize plant automation
- Programmed **200+** alarms, interlocks, and tracking logic based on **Cause & Effect Matrices** in DeltaV
- Mapped **1,000+** I/O points through bulk export/import using Excel

Product Development Intern

Amomii

May 2024 – Aug 2024

Taichung, Taiwan

- Leveraged **bilingual communication** (English/Mandarin) to facilitate interdisciplinary project coordination
- Programmed and documented Arduino modules, improving scalability and deployment efficiency by **25%**
- Facilitated **iterative design reviews** to align stakeholder feedback with integration priorities for home-users

District Energy Engineering Intern

Onni Group

May 2023 – Aug 2023

Vancouver, BC

- Evaluated renewable and low-carbon district energy and HVAC systems to drive sustainable building design
- Liaised with developers, contractors, and consultants to align mechanical design requirements
- Supported performance validation and testing to ensure compliance with energy efficiency goals

Mechanical Team Member

UBC Sustainingengineering

Jan 2023 – Sep 2023

Vancouver, BC

- Joined a 5-person sub team to construct a **small-scale wind turbine** to advance energy accessibility
- Modeled wind turbine pieces, such as ribs, base, and shaft by constructing CAD models using SOLIDWORKS

Projects

Capstone Project - IoT Structural Health Monitoring of Wind Turbines

Sep 2025 – April 2026

- Led real-time embedded firmware in C on ESP32 using **FreeRTOS** for multi-sensor **MEMS** acquisition
- Designed interrupt-driven data acquisition pipeline sustaining **99.83% data integrity** over 24-hour operation
- Implemented **autonomous** fault detection, logging, and recovery across network and sensor failures

MacBook Keyboard Chassis

Nov 2025 – Dec 2025

- Defined tooling, fixtures, datums, operation sequences in **Siemens NX** to machine thin-wall aluminum features
- Resolved non-manufacturable features by proposed geometry alternatives, achieving ± 0.05 mm tolerance

RC Vehicle Modification and Performance Optimization

Sep 2023 – Apr 2024

- Redesigned **chassis** using Glass Fiber Reinforced Polymer and **Resin Transfer Molding**, optimizing fiber orientation through tensile and bending tests, reducing weight by **15%**
- Optimized **shell** design via DFMA and FEA, enabling high-volume **thermoforming** (83,200 units/year)
- Engineered lightweight **tires and rims** using polyurethane and PVC via **injection molding**

Technical Skills

Programming Languages: Python, C, C#, MATLAB, ARM Assembly, PLC Programming (Ladder Logic), R

Software Tools: DeltaV Distributed Control System, SOLIDWORKS, Simulink, GitHub, Ansys, NX, ESP-IDF

Hands-on: Ethernet, SPI, I²C, PID Controller Tuning, Arduino, Oscilloscope, Multimeter, Function Generator

Languages: English (native), Mandarin Chinese (native)